

GRADUATION SHOWCASE

The Junior Innovator's Guide to Design Thinking

Welcome to the World of Design!

Did you know that you are already a designer? Every time you find a way to make something work better, you are using **Design Thinking** [cite: 70, 120].

What is a Gadget?

A **gadget** is more than just a piece of technology. It is a **tool that solves a problem** for people [cite: 73]. It can be as simple as a tool to pump air or as complex as a computer [cite: 74, 89].

Your Mission Today:

In this lesson, you will learn how real heroes use simple materials to solve big problems, and you will start your own journey using the **Design Loop** [cite: 82, 92].

Mini-Task: Look around right now. Spot 3 gadgets you use every day. Think about your water bottle, cycle pump, or lunchbox [cite: 74].

1. Functional Design

Design is not just about how things look—it's about how they work! We call this **Functional Design**: design made to serve a special purpose [cite: 86, 87].

Examples of Smart Design:

- **The Idli Stand:** A brilliant design that saves space by stacking plates on top of each other [cite: 88].
- **The Cycle Pump:** A functional tool that keeps your bike ready to ride whenever you need it [cite: 89].

2. Meet Real-Life Heroes

Meet two amazing Indian innovators who turned simple ideas into gadgets that changed lives [cite: 80, 81, 82]:

Mansukhbhai Prajapati

He invented the **Mitticool**—a fridge made entirely of clay! It keeps food and water cool without using any electricity [cite: 83].

Remya Jose

As a teenager, Remya created a **pedal-powered washing machine** to help her family wash clothes faster while also getting some exercise [cite: 84].

Both used **local materials** and clever ideas to solve everyday problems [cite: 85, 90].

3. The Design Loop: THINK

The **Design Loop** is a three-step process every innovator uses: **THINK** → **MAKE** → **SHARE** [cite: 92, 93].

STEP 1

THINK: Find the Problem

The first step is to look for a "bother." What is bothering you or someone you know? [cite: 95, 102, 103]

Case Study: Ravi's Water Bottle

Ravi noticed a problem: his water bottle always leaked in his school bag. This made his books wet and ruined his homework [cite: 104, 125].

Ravi's "Bother": Wet books caused by a leaky lid [cite: 126].

Your Turn!

Identify a problem you want to solve [cite: 105]. Write it down below:

What is the bother?

Who does it affect?

4. The Design Loop: MAKE

Now that you have identified a problem, it's time to build a solution! [cite: 96]

STEP 2

MAKE: Build Your Prototype

A **prototype** is your very first model. It doesn't have to be perfect—it just needs to show your idea [cite: 107, 108, 110].

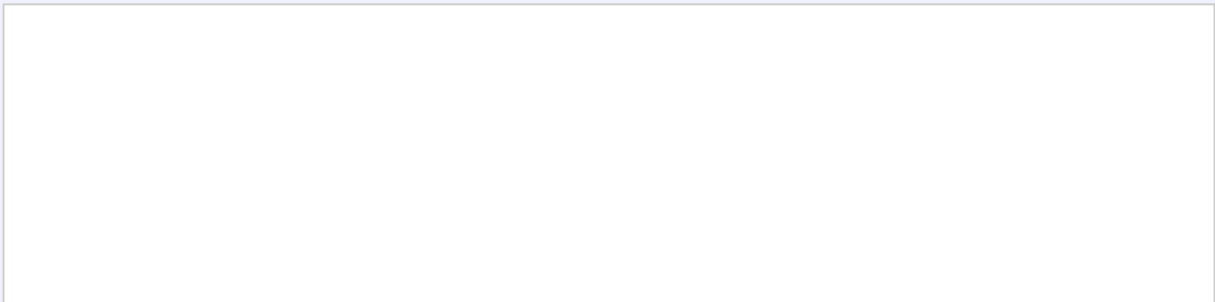
How to Build Your Prototype:

You don't need expensive tools. Use simple materials you can find around you [cite: 109]:

- **Cardboard & Paper:** For the body of your gadget.
- **Tape & Glue:** To stick parts together.
- **Recycled Items:** Bottle caps, old straws, or string.

Innovator Tip: Cut, fold, stick, and shape your materials. If it breaks or doesn't work the first time, don't worry! That is part of the loop. Just try a different way [cite: 110].

Design Sketch: In the space below, draw a quick sketch of what your prototype might look like.



5. The Design Loop: SHARE

An invention isn't finished until you share it with others! [cite: 97]

STEP 3

SHARE: Tell Your Story

Show your gadget to your friends, family, and classmates. Tell them how it helps solve the problem you found [cite: 112, 113].

Why Sharing Matters:

- **Get Better Ideas:** When others see your gadget, they might have great suggestions to improve it [cite: 114].
- **Inspire Others:** Your invention could inspire someone else to create something amazing too! [cite: 115]

Sharing Challenge:

Prepare a "30-second pitch" for your gadget. Explain:

1. What the problem was.
2. How your gadget solves it.
3. One thing you are proud of!

6. 3-2-1 Exit Review

Let's check what we've learned today to make sure you're ready to be an inventor [cite: 127, 129]!

3 Gadgets:

List 3 gadgets you learned about today (from the lesson or your surroundings) [cite: 130].

2 Steps:

Write down 2 steps of the **Design Loop** that you remember [cite: 131, 132].

1 Feeling Word:

What is one word that describes how you feel about being an innovator? [cite: 133]

Your Final Challenge:

Find one problem at home or school. Use the Design Loop to **THINK, MAKE, and SHARE** your gadget! Bring your idea or prototype to the next class [cite: 134, 135].

Thank you for being amazing designers! You are all inventors now!
[cite: 136]